

Lab 3: Component Modelling & Architectural Pattern Selection

Blood Bank Inventory & Emergency Donor Matcher (Problem Statement #12)

Architectural Choice

A three-layer style: a Presentation Layer (User Interface Component), a Business Layer (Inventory Manager, Emergency Request Manager, Donor Matching Service and Notification Service components) and a Data Layer (Database Component). The external SMS Gateway is reached only through the Notification Service. Components talk through provided/required interfaces such as IInventoryLedger, IDonorMatch, IAlertDispatch and ISMSGateway.

Two Reasons for the Choice

- 1. **Strict transactional consistency (NFR-001).** Every reservation passes through the Inventory Manager into one ACID inventory ledger, so two concurrent requests can never reserve the same blood bag. A distributed design would have to coordinate this across services and risk a patient-safety fault.
- 2. **Clear separation that maps directly to the requirements.** FR-002/003 live in the Inventory Manager, FR-004 in the Request Manager, FR-001/005 in the Donor Matching and Notification components. Each is replaceable behind its interface (for example, swapping the SMS provider), and the design suits a small team without microservice deployment overhead.

Security Advantage

Only the Business Layer may reach the Data Layer and the SMS Gateway, so the UI never touches donor phone numbers, medical data or the ledger directly. Role-based access (Emergency Requester vs Blood Bank Manager) and input validation are enforced at the business interfaces, and gateway credentials sit in a single component, the Notification Service.

Performance Benefit

Business components call each other in-process, avoiding the network hops of a microservice design. Cross-referencing inventory, finding donors within 10 km (indexed geo query) and dispatching SMS can therefore meet the FR-001 target of notifying compatible donors within 30 seconds. The stateless business layer can also be replicated behind a load balancer, with a replicated database, to support the 99.9% availability target (NFR-002)

